A missing plus in the original binary S4 display

Lean disproves the literal product parsing of the continuation lines in Semaev 2004, page 5, Remark 1. Inserting the missing addition gives an expression that Lean proves equal to the actual Sylvester resultant over every field of characteristic two, every a and every coordinate tuple. This concerns the printed display; the recursive construction is retained. [pcc:unverified]

Write s1=x+y+z+w, s3=xyz+xyw+xzw+yzw, s4=xyzw. The repair is:

```
s1^4 + s3^4 + s4*(s3+s1)^2 + s4^2*s1^2 + s3^2
```

The literal display multiplies its third and fourth terms. Over F_4 with $\text{alpha}^2+\text{alpha}+1=0$, the tuple $(1,1,\text{alpha},\text{alpha}^2)$ has s1=s3=s4=1. The actual resultant is zero; the literal product is one. All 16 tuples over F_2 agree with both expressions, so prime-field-only checks miss the issue. [pcc:unverified]

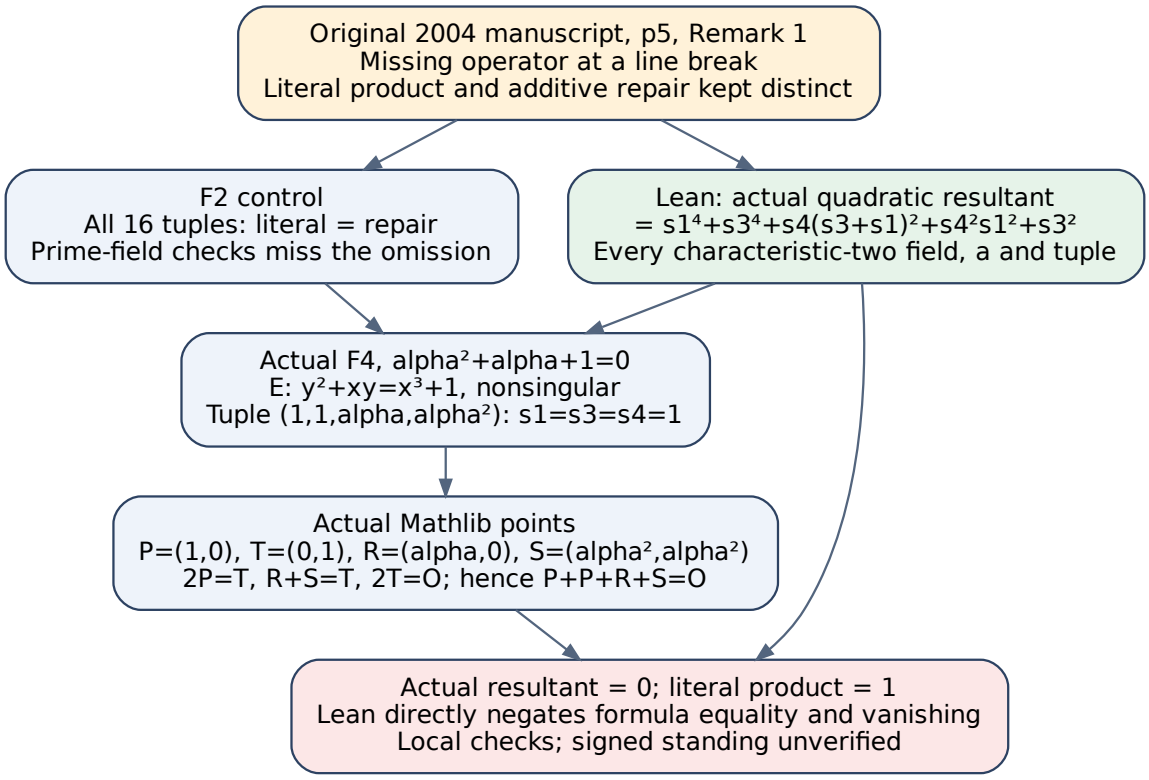


Figure 1: Universal repair, prime-field control and actual F_4 point witness. [pcc:unverified]

Actual points, proof map and manuscript read receipt

On the nonsingular actual curve E: $y^2+xy=x^3+1$ over F_4, Lean constructs P=(1,0), T=(0,1), R=(alpha,0) and S=(alpha^2,alpha^2). Actual Mathlib addition formulas give P+P=T, R+S=T and T+T=O. Hence P+P+R+S=O, while the literal display evaluates to one on their x-coordinates. The proof directly negates both universal formula equality and literal vanishing on actual four-point zero sums. It does not assume a lookup group or an algebraic-closure existence bridge in place of these rational points. [pcc:unverified]

Module / declarations	Exact checked obligation
`BinaryFourth`: `binary_resultant_coefficients`	Actual curve invariants and actual quadratic resultant coefficients
`BinaryFourth`: `actual_resultant_corrected`	Universal repaired formula, all characteristic-two fields, a and tuples
`BinaryFourthControl`	Actual F_4 cardinality/root relation; fixed symmetric quantities all one; resultant0/literal1
`BinaryFourthPoints`	Actual nonsingularity, four point equations, three actual additions and zero sum
`BinaryFourthCertificate`: `all_F2_tuples_agree`	Kernel check of all 16 prime-field tuples
Certificate: `literal_printed_recursion_claim_false`, `literal_printed_vanishing_claim_false`	Direct universal negations over actual F_4
Certificate: `binary_fourth_source_audit_certificate`	Universal repair plus actual counterexample and prime-field control

The main namespace is `SemaevBinaryFourthCertificateAudit`. The full gate checks 28 declarations, with zero failures and zero trusted-native exceptions. Lean 4.35.0 / Mathlib 56a00585 are pinned; only propext, Classical.choice and Quot.sound. No holes, custom axioms, native_decide, linter waivers or added elaboration settings. All 31 attempt files, including failed recursion, normalization and missing-import builds, are archived losslessly. Source commit `44b32e67` precedes theorem packet `PCC-20261110-95607f`. Actual local verdict and signed standing are retained in verification.json; signed standing remains unverified. [pcc:unverified]

The public author-uploaded original is dated 5 February 2004: six pages, 131064 bytes, SHA-256 1cab0ae675996cdba6393b1e369a22113fd2e74bcf1223595743dc140631c7d6. The entire manuscript, theorem proof, remarks and references were read; pages 2 and 5 were visually inspected. This is an additive current read receipt for S-SEMAEV-2004. Historical frozen plans still record their former failed IACR access. The typography claim is scoped to this exact PDF version, with no assertion about every later source or novelty. [pcc:unverified]

The original theorem uses algebraic-closure points; section 3 explicitly permits quadratic-extension y-coordinates. The earlier rational-only converse counterexample concerns an additional restriction. Existing all-arity geometric equivalence and individual-degree certificates stay unchanged. All-arity polynomial permutation invariance, absolute irreducibility, arbitrary-split exact normalization and qualified attack costs remain open obligations. No native experiment or paid compute. [pcc:unverified]

The byte-identical prior supplement IDC1hbf1f44625ee306dc covers the scalar six-coefficient resultant identity only. Its fixed 61-bit-prime evaluations cannot prove characteristic-two or actual-point claims; the new Lean proofs do so. KB tools/wrapper and Conductor unavailable; bounded primary, repository, Mathlib and formula/errata searches were used. No accepted correction source was found in that search scope. [pcc:unverified]

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